

Script Gerardo

Slide 1: Title

Today we are going to be presenting our experimental analysis on 3 different types of models to classify Natural disaster tweets.

Slide2: Motivation

In the age of social media, people instantly go to the web when something big happens. For example, we've all seen people pull out their phones to record a crash on their way to work and upload it online. Likewise, twitter offers people a fast and easy way to disseminate information for disasters such as floods, earthquakes and accidents which can be used to keep the public updated in real time.

However, people also use twitter to rant about their day and gossip. Such tweets contain a lot of disaster related words such as "crash", "flood" and "ablaze" that are used in slang that doesn't necessarily mean the tweet is talking about a real disaster.

Our Goal: To build a model that accurately discriminates between literal disaster reports and figurative language.

This system could be used to efficiently and accurately deploy first responders and resources to affected areas.

Slide 3: Example tweet

As can be seen, this tweet is talking about droughts in certain areas of the US. Accurately classifying this tweet could help deploy resources to the much needed areas.

Slide 4: Data set, Part I

Our data set comes from Kaggle, The original data only had a binary classification if the tweet was a disaster or not. We further classified the data set into 9 categories to train our models.

Slide 5: data set part II:

The 9 categories include: Accident/ Crash, Earthquake, Explosion, Fire/Wildfire, Flood, Not a disaster, Other disaster, Storm/Hurricane/Tornado, Violence/Attack/Terrorism

The data set was split 70/30 for training and testing respectively.

Slide 6: Data Quality:

As can be seen in the pie charts, the Train and Test have a very similar distribution between the different categories of tweets. Each category has roughly the same percentage of tweets with the exception of the two largest: Not a disaster and other disaster. The largest category is the "Not a disaster Category". Originally we had about 4,000 tweets for that category but we downsized it to create a better balance.